

Electrocatalytic Imprinted Polymer of N-Doped Hollow Carbon Nanosphere–Palladium Nanocomposite for Ultratrace Detection of Anticancer Drug 6-Mercaptopurine

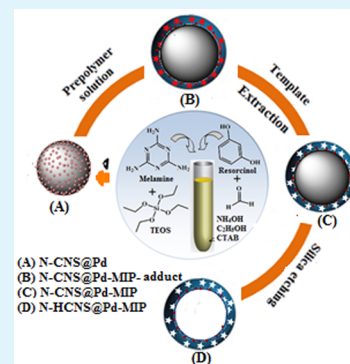
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Supporting Information

ABSTRACT: In this work, a nanohybrid-based imprinted polymer consisting of N-doped hollow carbon nanospheres and palladium is reported for the electroanalysis of ultratrace level of anticancer drug, 6-mercaptopurine, used in the treatment of leukemia. For this, N-doped carbon nanospheres decorated with palladium were first developed, and subsequently, a molecular imprinted polymer layer was grown onto their surfaces. The so-produced silica-embedded nanocomposite was made hollow by etching silica moieties with hydrofluoric acid. Finally, the whole system was doped on an ionic-liquid-modified pencil graphite electrode. The underlying synergistic effect of hollow carbon nanosphere-supported palladium nanoparticles inculcated electrocatalytic action. Notably, all rebinding sites in solid core–shells were confined within the shell, which hampers the effective diffusion of template. However, in this work, an effective diffusion of template across the hollow structure of inner and outer surfaces was observed. Consequently, this rendered approximately 2-fold heterogeneous rate constant as compared to the solid core–shell-based sensor. Differential pulse voltammetric transduction was used for ultratrace detection of 6-mercaptopurine through anodic stripping method. The hollow imprinted sensor revealed a linear dependence of current with concentration range 0.80–70.748 ng mL^{−1}. The limits of detection 0.11–0.22 ng mL^{−1} were realized in water, human blood plasma, urine, and pharmaceuticals. Thus, the proposed sensor demonstrated an attractive sensitivity reproducibility, as well as endurance requisite for the treatment of leukemia patients.

KEYWORDS: N-doped carbon nanospheres, hollow core–shell, molecularly imprinted polymer, 6-mercaptopurine, electrochemical sensor



INTRODUCTION

Molecularly imprinted polymers (MIPs) have selective recognition sites (molecular cavities) in synthetic polymeric receptors, having resemblance with structural formulations of template molecules. The hollow core–shell MIPs have aroused increasing attention in the field of sensing primarily because of their easy accessibility toward analyte binding in porous structure and favorable mass transport, as compared to the solid core–shells, with enhanced electron catalytic action.^{1–7} To the great surprise, very limited efforts have been made on hollow core–shell-based MIPs.^{8–12} However, these were mostly suffered from stability problem. Thus, the preparation of hollow carbon nanospheres for MIP development, possessing an inner mechanical support with a regular shape is necessarily required. Such system may facilitate fast transport of template during rebinding.⁹

Of various carbonaceous materials, carbon nanospheres (CNS) have increasing attention in MIP fabrication to render requisite functionalities for faster analyte absorption with excellent electrical conductivity and biocompatibility.^{13–17} The doping of N atoms into CNS texture proved advantageous to increase electrical conductivity, catalytic activity and stability of the system.^{18–23} This could be made feasible by either in

situ doping or post doping. Especially, in this work, the homogeneous incorporation of nitrogen into CNS was made using a nitrogen containing precursor (melamine), following an in situ silica assisted doping method. Accordingly, resin made of resorcinol and formaldehyde was used as a source of carbon, where melamine and tetraethoxysilane (TEOS) were source for nitrogen and structural support of system, respectively. Subsequently, a tubular furnace operating in inert condition was employed for the perfect carbonization to obtain N-CNS.

Electrocatalytic properties of N-CNS could be further improved by the support of Pd metal nanoparticles.^{24,25} Herein, the electronegative nitrogen atom having a lone pair of electrons on N-CNS acted as Lewis base to bind metal nanoparticles and provided an electrode catalytic N-CNS–Pd heterojunction.²⁰ This resulted in electronic activation and stable dispersion in the aqueous environment. For the first time, a spherical nanocomposite of N-CNS@Pd and MIP (N-CNS@Pd-MIP) was obtained in this work via precipitation

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polymerization. This required a suitable functional monomer, N-acryloyl derivative of aspartic acid (N-AAsp). The carboxylic functionalities of N-AAsp were utilized for anchoring metal ions. These core–shells of N-CNS@Pd-MIP composite were transformed into hollow N-CNS@Pd-MIP (N-HCNS@Pd-MIP) by the subsequent removal of embedded silica with the help of hydrofluoric acid. Despite the fact that the hollow CNS were successfully used in catalysis,^{26,27} lithium ion battery,^{28,29} oxygen-reduction,³⁰ supercapacitor,³¹ and water treatment,³² a very limited work on the application of HCNS in electrochemical sensor is reported.^{33,34} The N-HCNS@Pd-MIP prepared in this work was grafted on ionic liquid modified pencil graphite electrode (IL-PGE), following the “surface-grafting to” approach. Since, the selected monomer (N-AAsp), exists as a zwitterion³⁵ with hydrophilic behavior, it was interacted electrostatically with the cationic imidazolium ring of IL (1-butyl-3-methylimidazolium tetrafluoroborate)³⁶ dully immobilized on pencil graphite electrode (PGE). This offered a support material on PGE with effective electrochemical surface area (0.06 cm², Table 1). The sensor fabrication

Table 1. Different Parameters of Modified PGEs

electrodes	CV current (μ A) for 0.1 mM of [Fe(CN) ₆] ^{3-/4-}	ΔE_p (mV)	A_{eff} (cm ²)	R_f
IL-PGE	1.52	160	0.06	1.91
N-CNS/IL-PGE	5.72	140	0.17	5.41
N-CNS@Pd/IL-PGE	7.21	105	0.24	7.64
N-CNS@Pd-MIP/IL-PGE	9.68	95	0.74	23.56
N-HCNS@Pd-MIP/IL-PGE	13.78	80	1.16	36.94

protocol is mentioned as Scheme 1. Accordingly, the target 6-mercaptopurine (6-MP) was adsorbed in the molecular cavities of the hollow N-CNS@Pd-MIP at IL-PGE, primarily involving hydrogen bonding and electrostatic interactions. It was then desorbed (stripped off) anodically to register cyclic voltammetry (CV) and differential pulse anodic stripping voltammetry (DPASV) signals.

The model target, 6-MP, is a drug to treat leukemia, where it behaves as an antimetabolite to restrict the production of adenine nucleotides.³⁷ The 6-MP plasma concentrations are very difficult to maintain the required level of plasma³⁸ depending upon the weight of the patient. Thus, the decision of oral supplementation of 6-MP is a challenging issue. Several techniques related to high-performance liquid chromatography,³⁹ capillary electrophoresis,⁴⁰ spectrophotometry, and fluorescence⁴¹ were used for this. Nevertheless, one may not ignore the strenuous efforts in liquid chromatography for column and mobile phase selections. Also, the spectrofluorometry suffered from massive interferences. Although several electrochemical techniques are known for 6-MP detection, these are also lacked of selectivity and sensitivity.^{37,42–45} To date, only one attempt has been reported employing electrochemical MIP-based sensing.⁴⁶ However, this work too suffered from the problem of MIP-film stability and poor sensitivity. To overcome the above attendant problems including stability and sensitivity (see Results and Discussion, Interference and Endurance Aspects), a new N-HCNS@Pd-MIP/IL-PGE is proposed in the present instance, which revealed high endurance with no interferences and matrix effect.

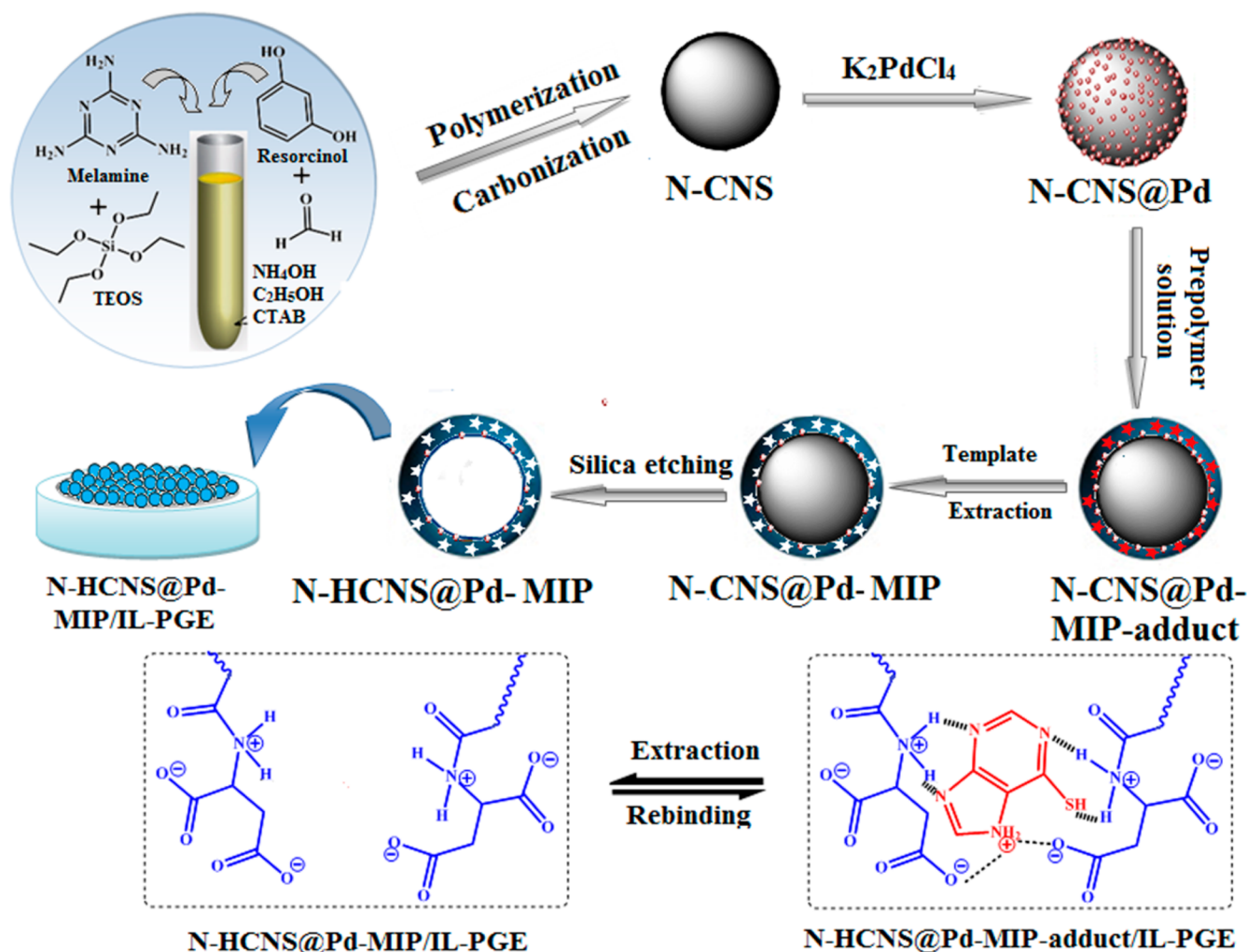
EXPERIMENTAL SECTION

Reagents and Apparatus. Resorcinol, formaldehyde solution (37–41%), melamine, aspartic acid, TEOS, cetyltrimethylammonium bromide (CTAB), ammonia aqueous solution, hydrofluoric acid (HF, 20%), ammonium persulfate (APS), N,N-methylene bis(acrylamide) (N,N-MBA), and acryloyl chloride (AC), have been obtained from Loba Chemie (Mumbai, India). The chemicals, IL, 6-MP, K₂PdCl₄ and associated interferents, namely, chlorambucil (CHB), ifosfamide (IFO), 5-fluorouracil (5-FU), 6-thioguanine (6-TG), ascorbic acid (AA), dopamine (DA), and glucose (Glu) were received from Sigma-Aldrich (Steinheim, Germany). Other interferents, namely, temozolomide (TMZ) and uric acid (UA) were obtained from Tokyo Chemical Industry, Co., Ltd. (Tokyo, Japan) and Loba Chemie (Mumbai, India), respectively. Different solvents like dichloromethane, tetrahydrofuran, chloroform, acetonitrile (ACN), triethylamine, acetone, ethyl acetate, and ethanol, used in this work, were used as such as obtained from Spectrochem Pvt., Ltd. (Mumbai, India). The supporting electrolyte, phosphate buffer (0.1 M) was used where its pH was adjusted to 5.0 with the help of drop of either HCl or NaOH. The demineralized triple distilled water (conductivity 0.06–0.07 $\times 10^{-6}$ S cm⁻¹) was used for the preparation of stock solution of 6-MP. The pharmaceutical formulation analyzed was 6-MP Tablet (Mercaptopurine, 50 mg/Tablet) as manufactured by Zydus Cadila healthcare Ltd. (Goa, India). The biological fluids (human blood plasma and urine) obtained from Institute of Medical Science, Banaras Hindu University, Varanasi, India and kept in ice pack before use. These samples were collected from stored stocks of pathology division, after the requisite ethical permissions of the University. The dilution of biological samples with water is must to minimize the complications from their matrices. The micropipettes (Top-Tech Biomedicals, Varanasi, India) were used for the delivery of all samples for experimentations.

Varian 3100 FT-IR (USA) spectrometer was used for the Fourier transform infrared spectroscopy (FT-IR) measurements, using pelletized form of sample in KBr. The morphology of dully immobilized PGE was studied using a high resolution scanning electron microscope (SEM, Nova Nanosem 450 FEI, Netherland) and atomic force microscope [AFM, NT-MDT Co., Russia]. For N-HCNS@Pd characterization, high-resolution transmission electron microscopic (TEM) analyses were performed on Technai-12FEI (Eindhoven, Netherlands). Accordingly, an EDAX detecting unit [Technai-20 (ST), 136-10, Port EDS, amplifier model 204] served effectively to obtain energy-dispersive X-ray spectra (EDS). CV and DPASV studies were carried out with the help of a portable potentiostat μ -Stat 200 (Drop Sens S.L. Oviedo, Spain). The three electrodes for this work were Ag/AgCl (3.0 M KCl) as reference, a platinum wire as counter and N-HCNS@Pd-MIP/IL-PGE (electrochemical surface area = 1.16 cm²) as working electrode. Herein, PGEs used for the modification of working electrode were made from pencil rods [2B, diameter 2 mm and length 5 cm, Hi Par, Camlin, Ltd. (Mumbai, India)] having bare surface area to the tune of 0.031 cm². For ascertaining surface area of MIP nanoparticles Brunauer–Emmett–Teller (BET) analysis was carried out with the help of micromeritics ASAP 2020 apparatus at –195.56 °C maintained by liquid nitrogen. The solid electrical conductivity was measured on a Keithley 236 Source Measure Unit. Ultrathin coatings of nanomaterials suspension over PGEs were ensured by a standard spin-coater (ACE $\rightarrow$ 200, Dong Ah Tech, Seoul, South Korea). X-ray photoelectron spectroscopy (XPS) experiments were done by PHI 5000 Versa Probe-II instrument (FEI Inc., USA). This instrument carries mono Al K α X-ray radiation (100.1 W, 100.0 μ , 45.0 °C, 187.85 eV) for the excitation. All analyses were made at 25 $\pm$ 1 °C.

Synthesis of N-AAsp. The functional monomer, N-AAsp was obtained on the basis of a known recipe.³⁵ Accordingly, AC (1.0 mL) was mixed in the solution of aspartic acid (1.33 g, 10 mL) and NaOH (0.4 g) in cold condition with continuous stirring. Its pH was kept constant between 7 and 8 with the help of 1.0 M NaOH. Subsequently, after 24 h reaction, the pH was brought down to $\sim$ 1.0 with the help of 1.0 M HCl. The monomer, N-AAsp, was

Scheme 1. Preparation of N-HCNS@Pd-MIP/IL-PGE



extracted with ethyl acetate and further dried with anhydrous sodium sulfate.

Synthesis of N-CNS@Pd. The N-CNS were synthesized following a known procedure.⁴⁷ Briefly, ammoniacal solution of CTAB (3.64 g, 120 mL aqueous ethanol) was heated with resorcinol (0.55 g), following with the addition of TEOS (3.29 mL) and formaldehyde (0.74 mL). After half an hour, a mixture of formaldehyde (0.55 mL) and melamine (0.3 g) solutions was introduced in above solution with continuous stirred for 24 h. This resulted in a residue which was collected by centrifugation and overnight drying at 70 °C. The final carbon-silica composites (N-CNS) were obtained by carbonization of as prepared solid product. The carbonization is accomplished under inert atmosphere in a tubular shape furnace at 800 °C (heating rate 1 °C min⁻¹ below 600 and 5 °C min⁻¹ above 600 °C). After carbonization in inert the atmosphere, the solid powder form of N-CNS were obtained. Herein, the perfect inert environment was ensured by the nitrogen purging in tubular furnace maintained ~800 °C (otherwise, all the components may turn to ash). For the decoration of Pd on the surface of N-CNS, 0.2 g of N-CNS and 0.05 g of K_2PdCl_4 (4:1, w/w, N-CNS and K_2PdCl_4 , optimized Figure S1C) were dispersed in 50 mL of distilled water in an ice bath with vigorous stirring.⁴⁸ After 30 min, obtained N-CNS@Pd were centrifuged and subsequently washed with water.

Fabrication of IL-PGE. For construction of PGE, 6 M HNO_3 was used for 15 min to treat pencil rod and then smoothened using soft cotton after drying the pencil rod. The pencil rod was inserted in a Teflon tube, where there could be microgaps in between pencil rod and tube. To fill up the gaps, a manual spray of epoxy resin was used.

The orifice of pencil rod was labeled by Emery paper (No. 400) and a metallic wire was employed at the reverse side for making electrical contact. Afterward, 10 μL of IL-solution (1:5, v/v, IL and water) was drop cast onto the labeled surface of PGE, involving physical adsorption and π - π attractions.

Fabrication of N-HCNS@Pd-MIP/IL-PGE. For MIP preparation, 10 mg of N-CNS@Pd hybrid was dispersed in phosphate buffer (pH 6.2), followed with ultrasonication. Afterward, 0.25 mmol of analyte (6-MP), 0.5 mmol of monomer (N-AAsp), and 2.5 mmol of cross-linking agent (N,N-MBA) were dissolved in 20 mL of phosphate buffer and then added into the sonicated N-CNS@Pd solution in inert condition. Precipitation polymerization was initiated by adding an initiator (250 μL APS, 10%, w/w) into the above reaction mixture, with stirring at 70 °C for 24 h. [The stoichiometry between template and monomer (1:2 molar ratio) and polymerization conditions were optimized as elaborated in section S.1 and Figure S1.] The obtained N-CNS@Pd-MIP was collected with the help of centrifugation, followed by template extraction with an extractant. Herein molecules of template were removed and MIP (template-free) particles were obtained, subjecting to water washings ($n = 3$, 2.0 mL each time) with the help of ACN solvent for 45 min. For this, the MIP-adduct (2.0 mg) was stirred in the solvent ACN (2.0 mL) and collected MIP-particles (template free) by the centrifugation of mixture. This was repeated for at least three times to ensure complete extraction of template molecules. The MIP nanomaterials were finally treated using methanol, followed by vacuum drying at 60 °C for overnight. The clear removal of 6-MP was ensured with the help of EDS analysis (Figure S2). Figure S2A displays elemental peaks (C, N, O, S and Pd)

corresponding to the MIP-adduct. Interestingly, the S peak related to the template was not observed after its complete removal, corroborating MIP structure (Figure S2B). For the preparation of hollow MIP nanomaterials (N-HCNS@Pd-MIP), the silica core of N-CNS@Pd-MIP was etched away by 20% HF solution for 1 h. Note that Pd NPs grown on the surface of N-CNS were not leached out with HF because the MIP layer onto N-CNS@Pd surface protected Pd NPs. In fact, the surrounding MIP coating here effectively prevented agglomeration. This restricted the possibility of Pd NPs leaching and consequently increased stability and electrocatalytic action of the system. The complete dissolution of only silica from N-CNS@Pd-MIP was reflected by EDS spectrum (Figure S2C). Such removal of silica by HF is also reported in a non-MIP system.⁴⁹ The nonimprinted nanomaterials (N-HCNS@Pd-NIP) were made following identical procedure, avoiding any template molecules.

An approach of “surface-grafting to” was utilized for N-HCNS@Pd-MIP doping onto IL-PGE. For this, a micropipette was employed to coat N-HCNS@Pd-MIP suspension (1.0 mg mL⁻¹ water) on the electrode. The N-HCNS@Pd-MIP was adsorbed on the IL-PGE via electrostatic interactions between positively charged IL-PGE (cationic imidazolium ring) and negatively charged N-HCNS@Pd-MIP. The N-HCNS@Pd-MIP-loaded IL-PGE was kept at room temperature after 30 min drying.

Electrochemical Measurements. The modified electrodes were used for electrochemical measurements using a supporting electrolyte of phosphate buffer (10 mL, pH 5.0). The analyte was added into the cell using standard addition method. The test solution was allowed to accumulate (E_{acc}) at -0.2 V for 90 s (accumulation time, t_{acc}), observing equilibration in 15 s (section S.2 and Figure S3). DPASV runs were taken in $+0.2$ to $+0.6$ V (vs Ag/AgCl) range at 10 mV s⁻¹, where amplitude of pulse was 25 mV with 15.0 Hz frequency. The nitrogen purging to eliminate oxygen from the cell was found unnecessary, since 6-MP oxidation was not interrupted. The CV measurements were performed in $+0.2$ to $+0.6$ V range, at different scan rates (20 – 200 mV s⁻¹). The limits of detection (LODs) were obtained as $S/N = 3$ where standard deviation (three times) of blank runs was divided by the slope of the calibration curve (6-MP concentrations versus current).

RESULTS AND DISCUSSION

Formation of N-CNS. Resorcinol and formaldehyde were polymerized in the presence of melamine in basic environment as reported earlier.⁵⁰ Accordingly, a monodispersed N-doped porous carbon nanosphere was prepared taking resorcinol and formaldehyde (as a carbon precursors), TEOS (as silica precursor), and melamine (as a nitrogen source) (Scheme S1). In this preparation, TEOS underwent hydrolysis to form silicic acid, which gradually converted after condensation to form an oligomer of TEOS. When Si number is higher than 3, the oligomer preferred to cyclization (ring structure), rather assuming linear-chained oligomerization.⁵¹ The condensed oligomers continued to grow until silica particles were formed. The silica nanoparticles so-produced was reacted with the $-OH$ groups of condensed resorcinol-formaldehyde species. The addition of CTAB (surfactant) latter formed polymer–silica-surfactant composite spheres. The melamine and formaldehyde were finally added to form nitrogen containing polymer–silica-surfactant composite. This eventually co-condensed to form branched polymeric species. The subsequent carbonization in inert atmosphere resulted in N-CNS. The carbonization actually helped to alter the geometry of spheres in the composite to yield porous core–shell spheres.

Spectral Characterizations of N-CNS and N-CNS@Pd. The formation of N-CNS is supported by analyzing XPS spectrum (Figure 1A). The high-resolution curve-fitted XPS spectrum of N 1s shows two peaks. These peaks correspond to

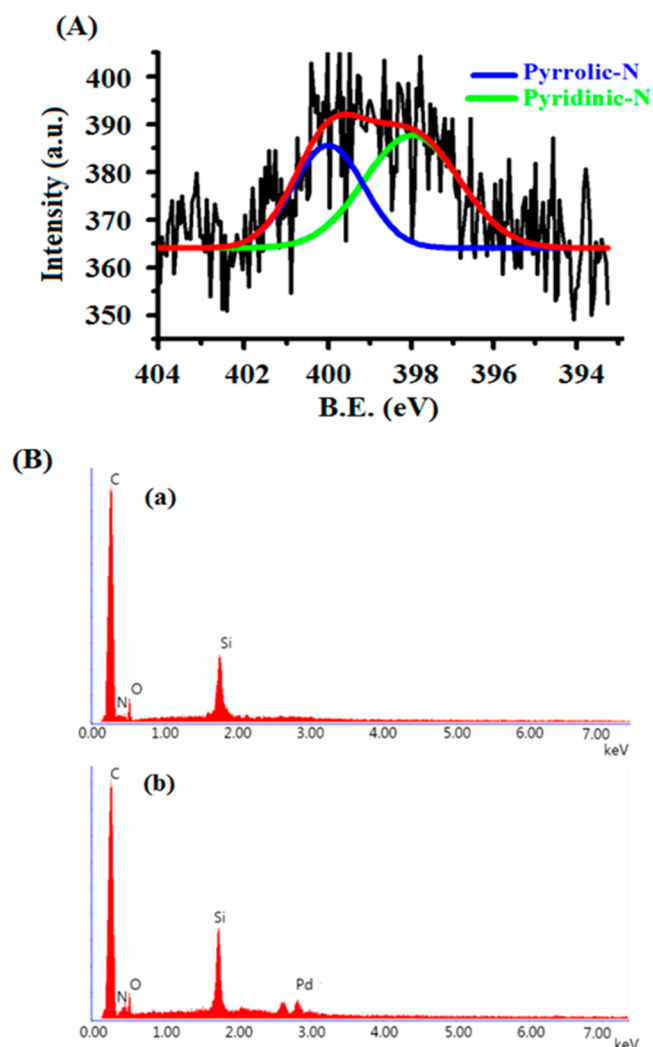


Figure 1. (A) High-resolution curve-fitted N 1s XPS-spectrum of N-CNS and (B) EDS spectra of N-CNS (a) and N-CNS@Pd (b).

the nitrogen functionalities of pyridinic (N-6, 397.89 eV) and pyrrolic groups (N-5, 399.8 eV). This supported N-doping in the carbon network of N-CNS.⁴⁷ The introduction of Pd NPs on N-CNS is supported by EDS analysis (Figure 1B). Accordingly, the EDS spectrum of N-CNS (curve a) shows peaks for C, N, O, and Si. After Pd deposition on N-CNS, an additional peak for Pd is found in the spectrum (curve b). Lewis base property (lone pair electrons) of nitrogen atoms in N-CNS surface could be accounted for the active anchoring of Pd NPs, through the formation of uniform dispersion to realize an effective for electronic activation in aqueous media.

Electrodiscs of the Modified PGEs. CV runs of redox probe (ferricyanide, 0.1 mM) at different modified PGEs are recorded in the supporting electrolyte at 100 mVs⁻¹ (Figure S4). The peak potential difference (ΔE_p) attributes the difference in potentials of anodic and cathodic waves ($\Delta E_p = E_{pa} - E_{pc}$). The ΔE_p values at IL-PGE (curve a), N-CNS/IL-PGE (curve b), N-CNS@Pd/IL-PGE (curve c), N-CNS@Pd-MIP/IL-PGE (curve d), and N-HCNS@Pd-MIP/IL-PGE (curve e) are found to be 160, 140, 105, 95, and 80 mV, respectively. As can be seen, the lowest value of ΔE_p (80 mV) signified perfect reversibility at N-HCNS@Pd-MIP/IL-PGE. This might be attributed a synergism between MIP nanocomposite and N-HCNS@Pd hybrid, where the electro-

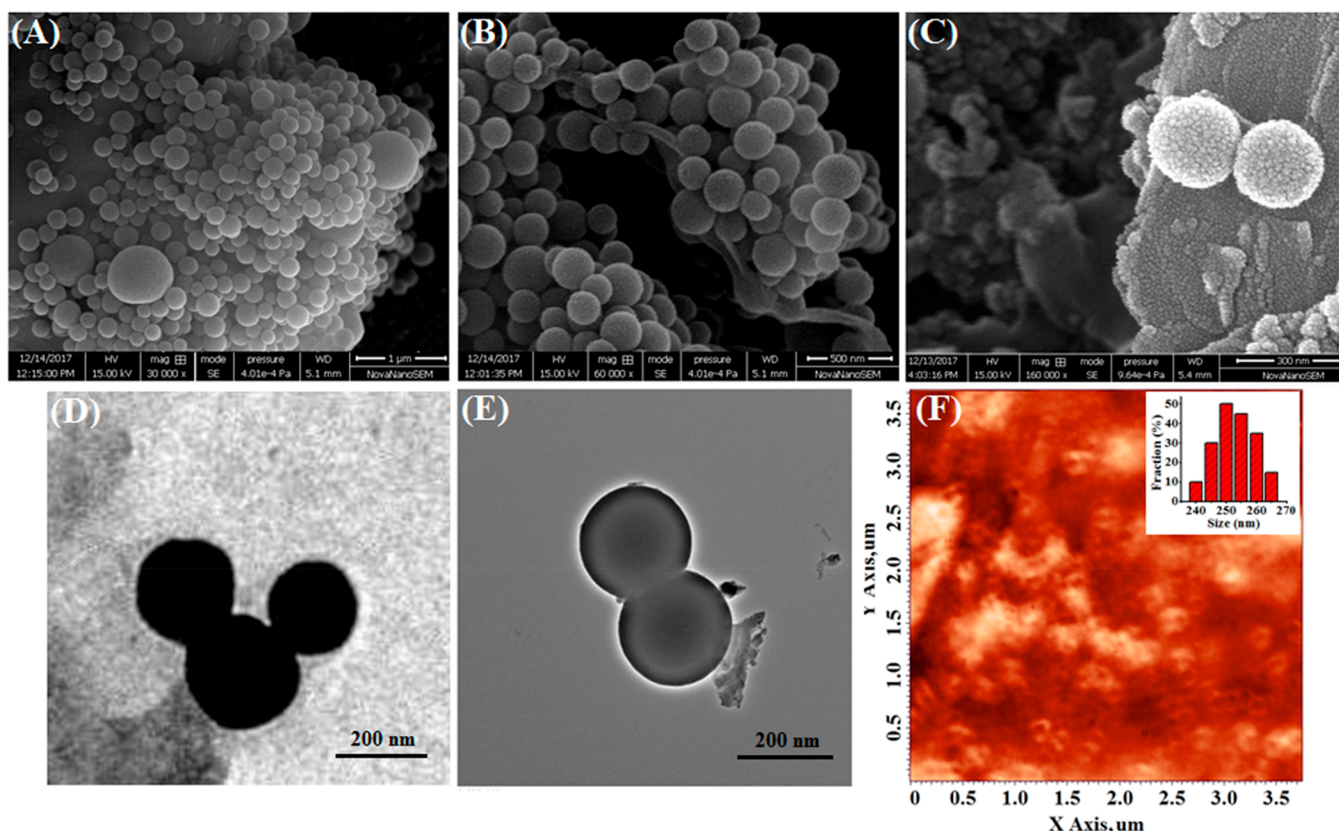


Figure 2. SEM images of (A) N-CNS, (B) N-CNS@Pd-MIP, and (C) N-HCNS@Pd-MIP. TEM images of (D) N-CNS@Pd-MIP and (E) N-HCNS@Pd-MIP. (F) AFM image of N-HCNS@Pd-MIP. (Inset: Histogram representing N-HCNS@Pd-MIP particles size distribution.)

catalytic action of N-CNS and Pd heterojunction drastically reduced the overpotential of electron transfer process. Furthermore, CV responses are gradually increased from IL-PGE to N-HCNS@Pd-MIP/IL-PGE with increasing effective surface area (A_{eff}) and roughness factor (R_f) (Table 1). The maximum current response is obtained for N-HCNS@Pd-MIP/IL-PGE. It could be due to easy accessibility, favorable mass transport, and improving binding capacities, which facilitated diffusion of probe molecule toward the recognition sites. Herein, the A_{eff} values could be obtained using reversible process represented by Randles–Sevcik equation⁵²

$$I_p = (2.687 \times 10^5) n^{3/2} \nu^{1/2} D^{1/2} A C^0 \quad (1)$$

where I_p is the anodic peak current, D the diffusion coefficient of ferricyanide probe ($0.76 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$),⁵³ n the number of electrons $\nu^{1/2}$, the square root of scan rate, A the effective surface area, and C^0 corresponds to concentration of the probe solution. The R_f values are calculated by dividing A_{eff} with A_{geom} [geometrical surface area (0.031 cm^2) of bare PGE]. Therefore, the relative high values of A_{eff} and R_f of N-HCNS@Pd-MIP, favored a very sensitive and selective detection of the analyte.

Spectroscopy and Morphology. The analyte binding in molecular cavities involved hydrogen bonding and electrostatic interactions (Scheme 1). This is revealed by FT-IR (KBr) spectra (section S.3 and Figure S5).

Surface morphologies of the various particles (N-CNS, N-CNS@Pd-MIP, and N-HCNS@Pd-MIP) at IL-PGEs are investigated through SEM images. It is clearly visualized that the synthesized N-CNS has uniform spherical size distribution with an average diameter $\sim 178 \text{ nm}$ (Figure 2A). Upon MIP

fabrication on N-CNS@Pd hybrid, a bunch of grapes like morphology is appeared with an average size $\sim 240 \text{ nm}$ for each particle (Figure 2B). The hollow morphology (porous) of N-HCNS@Pd-MIP is observed at magnified scale because of HF-etching of silica from the core of N-CNS@Pd-MIP nanocomposite (Figure 2C). Surface morphologies of solid and hollow MIP nanospheres are further differentiated from TEM images, with a uniform size distribution ($246 \pm 10 \text{ nm}$ per particle) (Figure 2D and E). The morphology of N-HCNS@Pd-MIP is also investigated by AFM analysis (Figure 2F). It reflects the hollow core-shell structured MIP nanomaterials with uniform size distribution $250 \pm 15 \text{ nm}$ per particle (Figure 2F, Inset).

Measurements of Surface Properties. N-CNS@Pd-MIP and N-HCNS@Pd-MIP were analyzed to calculate surface area by nitrogen adsorption in BET techniques. The results including water swelling of the systems (specific surface areas 117.62 and $136.17 \text{ m}^2 \text{ g}^{-1}$ with swelling ratios 38.6% and 53.4% for N-CNS@Pd-MIP and N-HCNS@Pd-MIP, respectively) are reported in section S.4. Accordingly, the hollow core-shell texture of MIP increases the specific surface area and swelling characteristics than those for corresponding solid core-shell MIP. Conductivities of N-CNS@Pd-MIP and N-HCNS@Pd-MIP were also calculated with the help of two-probe method. The N-HCNS@Pd-MIP revealed conductivity to the almost 10-fold higher value with respect to N-CNS@Pd-MIP. This may be presumably because of elution of silica from the MIP core, which led to formation of more conducting N-HCNS@Pd-MIP (section S.5).

Electrochemistry and Adsorption Behavior. Figure 3A displays CV (anodic stripping) responses of 6-MP (32.88 ng

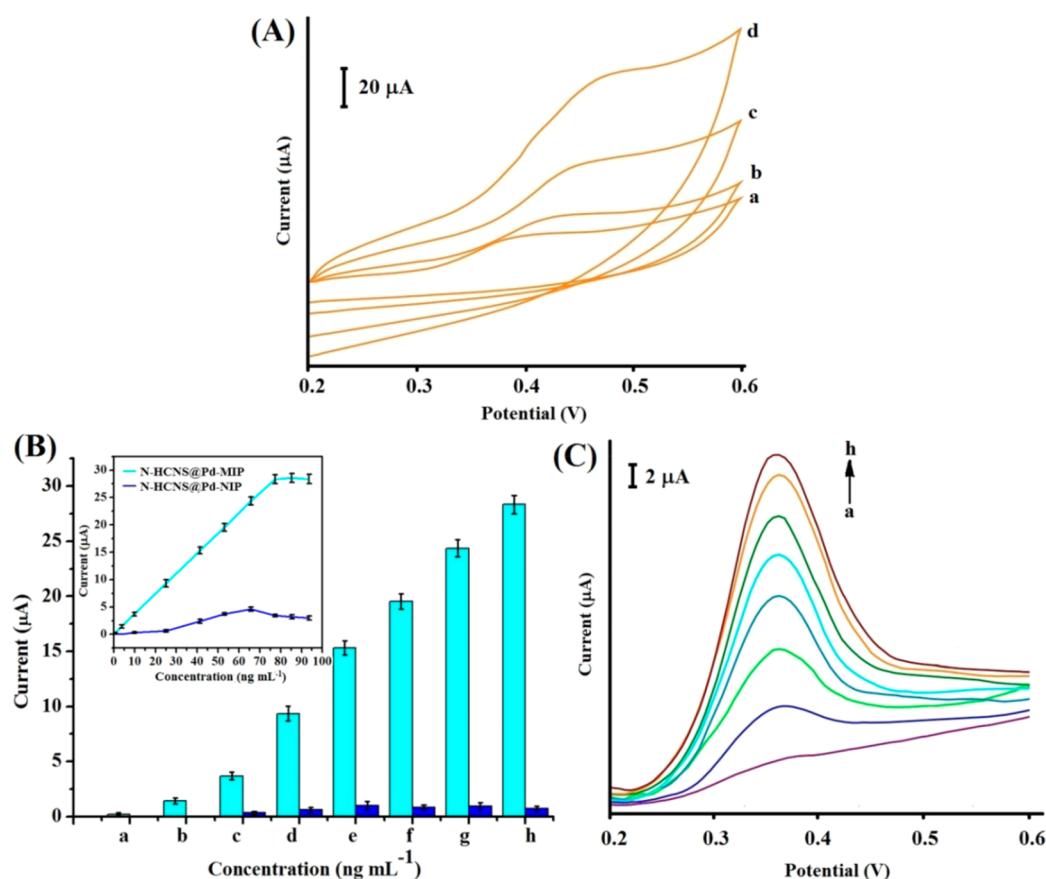


Figure 3. (A) CV responses on N-HCNS@Pd-MIP/IL-PGE for 6-MP (32.8 ng mL^{-1}) at different scan rates: (a) 20, (b) 50, (c) 100, and (d) 200 mV S^{-1} . (B) DPASV signals on N-HCNS@Pd-MIP/IL-PGE and N-HCNS@Pd-NIP/IL-PGE for different 6-MP concentration in aqueous environment (a–h): 0.63, 3.82, 9.89, 25.03, 41.26, 52.98, 65.71, and 77.31 ng mL^{-1} . (C) DPASV representative curve of different 6-MP concentrations (from a to h) in blood plasma sample: 0.80, 12.47, 22.96, 36.72, 52.25, 64.89, and 70.75 ng mL^{-1} .

mL^{-1}) at $20\text{--}200 \text{ mV s}^{-1}$ scan rates on N-HCNS@Pd-MIP/IL-PGE. Accordingly, irreversibility was observed to have positive shifting of peak current with scan rate and could be accorded to the high energy for electro-oxidation in the limited time span at higher scan rates. The 6-MP oxidation involved the participation of protons as evident from the negative shift of current with pH ($2.2\text{--}10.0$) variation in accordance with the following equation:

$$E_p(\text{V}) = (-0.055 \pm 0.006)\text{pH} + (0.664 \pm 0.042), \quad (R^2 = 0.97) \quad (2)$$

Herein, the slope (0.055 V/pH) of E_p versus pH profile is close to the value from the Nernst equation (0.059). This suggested the equal number of protons and electrons participated in oxidation of 6-MP. It corroborates the mechanism of 6-MP electro-oxidation as earlier reported involving $2e^-$ and $2H^+$.⁴² In the earlier work,⁴² CTAB-modified electrode required a high overpotential (0.45 V) for electro-oxidation of 6-MP. This caused electrode fouling and massive interferences by other electroactive endogenous species. However, in the present case, the proposed sensor effectively reduced the oxidation overpotential of 6-MP to the tune of 0.36 V by virtue of improved electrocatalytic activity. Such improvement in electrocatalytic property of N-HCNS@Pd-MIP nanocomposite might be attributed to the synergistic contribution of Pd NPs and N-HCNS. In fact, Pd NPs served as noble metal electro-catalysts for 6-MP oxidation. The N-HCNS scaffold

enabled the higher adsorption of Pd NPs which paved the way for efficient electrochemical oxidation. The CV responses depicted a linear behavior between E_{pa} vs $\log \nu$ and I_{pa} (anodic peak current) vs $\nu^{1/2}$ on N-HCNS@Pd-MIP/IL-PGE:

$$E_{pa}(\text{V}) = (0.097 \pm 0.008)\log \nu + (0.539 \pm 0.01), \quad (R^2 = 0.98) \quad (3)$$

$$I_{pa}(\mu\text{A}) = (41.99 \pm 4.209)\nu^{1/2} + (0.783 \pm 1.29), \quad (R^2 = 0.98) \quad (4)$$

This supported the involvement of a diffusion-controlled process in electro-oxidation of 6-MP. Furthermore, the test analyte showed irreversibility in accordance with

$$E_p = E^{0'} + \frac{RT}{(1-\alpha)nF} \ln \nu \quad (5)$$

where E_p is the potential for anodic wave, $E^{0'}$ the formal redox potential, α the electron transfer coefficient, n the number of electron transfer, ν the scan rate, F the Faraday's constant, R the gas constant, and T is temperature. If one realizes α to be fractional, the electron transfer can be considered of irreversible nature. This value was observed to be 0.69 , in the support of irreversible oxidation of 6-MP. The higher electro-kinetic process involved N-CNS@Pd, improving heterogeneous electron transfer rate constants (k). These values were 5.06×10^{-3} and $9.62 \times 10^{-3} \text{ cm s}^{-1}$ at N-CNS@

Table 2. DPASV Responses of 6-MP in Aqueous, Real, and Pharmaceutical Samples

sample	regression equation	range (ng mL ⁻¹)	recovery ^a (%)	LOD ^b (3σ) (ng mL ⁻¹)	RSD ^c (%) (n = 3)
aqueous solution	$I_p = (0.368 \pm 0.001)C + (0.054 \pm 0.057)$, $n = 8$, $R^2 = 0.99$	0.63–77.31	97.4–101.0	0.11	0.28
blood plasma (diluted 100 times)	$I_p = (0.370 \pm 0.001)C + (0.044 \pm 0.042)$, $n = 7$, $R^2 = 0.98$	0.80–70.75	98.4–100.4	0.15	0.23
urine (diluted 100 times)	$I_p = (0.366 \pm 0.0007)C + (0.024 \pm 0.034)$, $n = 6$, $R^2 = 0.99$	0.74–74.85	97.5–99.8	0.22	0.21
pharmaceutical ^d	$I_p = (0.361 \pm 0.001)C + (0.164 \pm 0.100)$, $n = 7$, $R^2 = 0.99$	0.65–80.92	97.7–100.8	0.16	0.27

^a% Recovery = (amount of analyte determined/amount of analyte taken) × 100. ^bLOD based on the minimum distinguishable signal for lower concentrations of analyte ($S/N = 3$, confidence level 95%). ^cRSD (%) for three sets of LOD data. ^dDiluted 7.74×10^6 times, endogenous concentration is 4.9×10^6 ng mL⁻¹ (certified values 5.0×10^6 ng mL⁻¹).

Table 3. Comparative Study of 6-MP at Different Electrodes/Methods^a

sample no.	electrodes/methods	range (ng mL ⁻¹)	LOD (ng mL ⁻¹)	remarks	refs
1	MWCNTPE	$85.09\text{--}15.31 \times 10^4$	17.01	no interference study, poor sensitivity, no blood plasma analysis	37
2	fluorescence and spectrophotometric	$6.80\text{--}20.42 \times 10^2$ and $13.61\text{--}544.60$	1.70 and 3.40	expensive instrumentation, poor sensitivity, no analysis in urine, and pharmaceutical samples	41
3	MWCNT-CTAB/GC	76.08–608.68	1.21	poor sensitivity and selectivity, no analysis in pharmaceuticals	42
4	<i>p</i> -APMCNTPE-TiO ₂	$15.31\text{--}59.56 \times 10^3$	11.06	poor sensitivity and selectivity, no blood plasma analysis	43
5	PP/MWCNTs/PGE	$34.03\text{--}170.19 \times 10^2$	13.61	no interference study, poor sensitivity	44
6	sol-gel/ZnO@GQDs MIP PGE	$0.84\text{--}7.60 \times 10^3$	1.52	poor stability and sensitivity, involved much time for sensor fabrication	46
7	N-HCNS@Pd-MIP/IL-PGE	0.63–77.31	0.11	analyses in blood plasma, urine and pharmaceutical samples, studied interferents, sensor fabrication easier with faster analysis (~2.0 min)	present work

^aMWCNTPE: Isoprenaline-modified multiwall carbon nanotubes paste electrode. CTAB/GC: Cetyltrimethylammonium bromide/glassy carbon electrode. *p*-APMCNTPE-TiO₂: *p*-Aminophenol modified carbon nanotubes electrode. PP: Polypyrrole. GQDs: Graphene quantum dots.

Pd-MIP/IL-PGE and N-HCNS@Pd-MIP/IL-PGE, respectively (section S.6). The maximum value of k at N-HCNS@Pd-MIP/IL-PGE might be accorded to the hollow core–shell structure of MIP, where template binding was enhanced and mass transport was largely improved in comparison to the solid core–shell MIP.

Additional chronocoulometric experiments were carried out to reveal the kinetics of electrode (section S.6). This showed the surface coverage (Γ^0) to be 5.38×10^{-12} and 1.99×10^{-11} mol cm⁻² and D values 1.04×10^{-4} and 3.76×10^{-4} cm² s⁻¹ at N-HCNS@Pd-MIP/IL-PGE and N-HCNS@Pd-MIP/IL-PGE, respectively. The higher D value at N-HCNS@Pd-MIP/IL-PGE was due to its larger effective surface area with enhanced binding, compared to that with the solid core–shells MIP. Such hollow structure in nanospheres of N-HCNS@Pd-MIP might provide a shorter path in larger electro-active region for analyte diffusion. Further, it may be best fitted in the linear Langmuir adsorption isotherm, which reveals specific adsorptions via H-bondings and electrostatic attractions in the monolayer of hollow core–shell MIP. The chronocoulometric data in this work revealed free energy change (ΔG) depending upon adsorption coefficient (b) as $\Delta G = -RT \ln b$. Accordingly, ΔG and b were estimated as -58.56 kJ mol⁻¹ and 1.83×10^{10} L mol⁻¹, respectively, indicating the spontaneous inclusion of analyte molecules (section S.7).

Practical Applications. Figure 3B describes the signals of DPASV in water for 6-MP ($0.63\text{--}77.31$ ng mL⁻¹), at MIP and NIP modified IL-PGEs. Herein, the peak current at N-HCNS@Pd-MIP/IL-PGE was linearly increased with the increasing concentrations of the template (Figure 3B inset). This showed selective determination in the specific molecular cavities of imprinted polymer. Contrarily, the DPASV peak

currents at N-HCNS@Pd-MIP/IL-PGE have not shown any linear variation (Figure 3B inset) presumably due to the absence of tailor-made selective recognition sites and random absorption of analyte on the nonimprinted matrix. Fortunately, the nonspecific adsorption of 6-MP at NIP modified IL-PGE was completely mitigated by multiple water-washings ($n = 3$, 3.0 mL). As a safe-guard, MIP modified electrode was also subjected to similar water-washing to eliminate the nonspecific adsorption, if any. Blood plasma, urine, and pharmaceutical samples were examined following the standard addition protocol. To largely mitigate the matrix effect, the water dilutions of these real samples was necessarily required. The biological fluids were 100 times dilute, whereas the pharmaceutical sample was subjected to dilution of 7.74×10^6 times with water. Also such dilution actually helped to move the analysis in the required ranges. Table 2 depicts linear equations (I_p vs C), correlation coefficients (R^2), recoveries, and corresponding LODs. The LODs of 6-MP having concentration range of $0.63\text{--}80.92$ ng mL⁻¹ were realized very low as $0.11\text{--}0.22$ ng mL⁻¹. Although the optimal blood plasma concentration of 6-MP is reportedly $152.17\text{--}15.17 \times 10^2$ ng mL⁻¹, this amount depletes in chronic cases of 6-MP deficiency in Leukemic patients.⁵⁴ The proposed sensor, which is effective for the 6-MP detection up to ultratrace level in real samples, can be considered useful in drug supplementation. Further, the experimented recovery of 6-MP in the real samples was obtained to be quantitative (97.5–100.8%), with absolutely no cross-selectivity and complications of sample matrices. A complete saturation of binding sites was noticed beyond 70.75 ng mL⁻¹ in blood plasma sample, as revealed by DPASV measurements (Figure 3C).

The DPASV peaks of 6-MP in blood plasma have symmetrical features. Such peaks, symmetrical in nature, were also observed with other real samples. One may conclude these symmetrical peaks are akin to water sample presumably owing to the massive dilution of real samples with water.

It is worth comparing N-HCNS@Pd-MIP/IL-PGE with reported electrodes/methods of 6-MP analysis (Table 3). The reported known electrodes/methods lacked sensitivity of the measurement. Some methods are reported without studying interference. The present sensing method is relatively ultra-sensitive and found to be applicable in real-time analysis for 6-MP, without sample complications. Where assay time is concerned, Table 3 demonstrated the proposed sensor to be advantageous in clinical setting, which is relatively fast taking predictably 2 min for electrochemical measurements, as compared to the other sol-gel, MIP-based sensors.⁴⁶

Interference and Endurance Aspects. The analyte, 6-MP (30.0 ng mL⁻¹), individually and with some anticancer drugs, such as CHB, IFO, TMZ, 5-FU, 6-TG, and biomolecules AA, DA, Glu, and UA (taken in 1:200 molar ratios, 6×10^3 ng mL⁻¹ each), were investigated on N-HCNS@Pd-MIP/IL-PGE (or NIP) modified electrodes (Table S1). Individual analysis of interferents showed frivolous response. On the other hand, N-HCNS@Pd-MIP/IL-PGE response was very quantitative (100%) for 6-MP, on account of effective imprinting. In individual capacity, all interferents have shown some DPASV currents, without subjecting them for water-washings; multiple water-washings fortuitously removed current of all interferents (Figure 4). In this context,

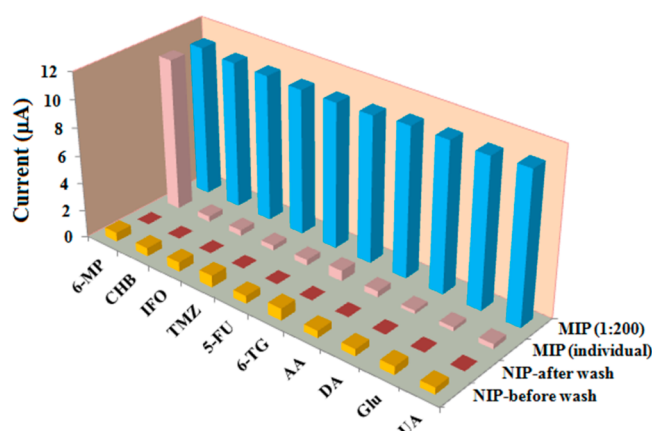


Figure 4. Analyte and interferent response in individual capacities and in 1:200 binary mixtures on N-HCNS@Pd-MIP and N-HCNS@Pd-NIP-modified IL-PGEs.

selectivity coefficients (k), relative selectivity coefficients (k') and imprinting factor ($IF = i_{MIP}/i_{NIP}$) were calculated (Table S1). The IF values were obtained for 6-MP, CHB, IFO, TMZ, 5-FU, 6-TG, AA, DA, Glu, and UA as 17.25, 0.66, 0.617, 0.428, 0.750, 0.813, 0.846, 0.600, 0.466, and 0.750, respectively. The IF (17.25) for 6-MP in this work revealed 20 times higher value than that for interferences.

The reproducibility of the proposed sensor was investigated by measuring DPASV responses of 6-MP (30.0 ng mL⁻¹) using five electrodes prepared under the identical conditions. As a result, the peak current 1.30% varying with relative standard deviation (RSD), which indicated a good reproducibility in terms of electrode-to-electrode variations. Moreover, the reproducibility of a single sensor was examined for 15

successive runs; the RSD obtained was 1.11%. After every run, the sensor has to be regenerated by removing bound template with extractant (ACN) for the next use. The long-term stability of the sensor, which depends on both the stability of the MIP-film and the activity of the 6-MP, is also important to explore its practicability. The sensor response actually showed decreasing tendency by only 5% as compared with its initial response, after 3 weeks duration. The modified electrodes are stable for 3 weeks if stored at the room temperature in dark condition.

CONCLUSION

We have developed an electrochemical sensor based on N-HCNS@Pd-MIP where unique advantages of synergism and electrocatalytic activity were taken into accounts. It was subjected in clinical settings for ultratrace analysis of 6-MP. N-HCNS@Pd incorporated MIP actually behaved as a suitable material for the enhancement of electrocatalytic activity, stability, and sensitivity. As much as, 2-fold heterogeneous rate constant, 3.5-fold diffusion coefficient, and 10-fold electrical conductivity were obtained on N-HCNS@Pd-MIP as compared to N-CNS@Pd-MIP decorated sensor. The proposed sensor, N-HCNS@Pd-MIP exhibited a good degree of reproducibility, stability, and anti-interference ability to decide adequate drug supplementation toward chemotherapy of leukemia patients.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.9b02947.

Schematic representation for the formation of N-CNS, optimization conditions for polymerization, EDS spectra, optimization of analytical parameters, CV runs of potassium ferricyanide probe at various modified electrodes, FT-IR spectra of analyte, functional monomer, MIP-adduct, and MIP, and selectivity coefficients, relative selectivity coefficients, and imprinting factor (PDF)

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Notes

The authors declare no competing financial interest.

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